

# Sleep Efficiency Classification Using Machine Learning

Iowa State University

COM S 474 Project Proposal

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# 1 Project Title

Sleep Efficiency Classification Using Machine Learning

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## 3 Project Details

### 3.1 Project Objective

The objective of this project is to use machine learning algorithms to predict whether a person has good or poor sleep efficiency based on their lifestyle habits and health information. The problem we are solving is determining how factors such as caffeine consumption, alcohol intake, smoking status, exercise frequency, and sleep composition affect overall sleep efficiency. Quality sleep is extremely crucial to maintaining a healthy mind and body. Understanding how our everyday activities affect our rest would be beneficial and could help people make better lifestyle decisions.

Machine learning can help solve this problem because there are many interacting variables that influence sleep efficiency. Instead of analyzing each factor individually, classification algorithms can learn patterns from the data and determine which features are most important in predicting sleep efficiency. Additionally, machine learning allows us to compare different models and evaluate which one performs best on this dataset.

### 3.2 Datasets

We are using the Sleep and Lifestyle Health dataset found on Kaggle. This is a CSV dataset constructed with 1000 rows and 15 columns, where each column represents a different feature that factors into one's sleep length and quality. This is a synthetic dataset, but it simulates the effects of different habits and health variables on sleep. The columns and their descriptions can be observed in the table below.

| Feature | Description                             |
|---------|-----------------------------------------|
| ID      | Database identifying integer for sample |

|                      |                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------|
| Age                  | Age of the sleeping person                                                                    |
| Gender               | Biological sex of the person                                                                  |
| Bedtime              | Time the person goes to bed, in HH:MM format and in 24 hour time                              |
| WakeUpTime           | Time the person wakes up, in HH:MM format                                                     |
| SleepDuration        | Total hours slept between bedtime and wake-up time.                                           |
| SleepEfficiency      | Percentage of time spent asleep while in bed. Formula is (sleep duration / time in bed) * 100 |
| REMSleepPercentage   | Percentage of total sleep spent in Rapid Eye Movement (REM) phase.                            |
| DeepSleepPercentage  | Percentage of total sleep spent in deep sleep.                                                |
| LightSleepPercentage | Percentage of total sleep spent in light sleep.                                               |
| Awakenings           | The number of times woken up during sleep.                                                    |
| CaffeineConsumption  | Caffeine consumption in mg consumed.                                                          |
| AlcoholConsumption   | Alcohol consumption in number of standard drinks.                                             |
| SmokingStatus        | Boolean for if the person is a smoker.                                                        |
| ExerciseFrequency    | How many times the person exercises per week.                                                 |

For this project, we will convert the SleepEfficiency percentage into 2 categories:

- Good Sleep Efficiency ( $\geq 85\%$ )
- Poor Sleep Efficiency ( $< 85\%$ )

This allows us to use supervised classification algorithms. The ID column will only be used as an identifier and not as a predictive feature.

This dataset will be split into training and testing data. There are 1000 total samples, so a reasonable split would vary around this. Training: 700 samples (70%), Validation: 150 samples (15%), and Testing: 150 samples (15%).

### 3.3 Machine Learning Algorithm

We will implement and compare three classification algorithms:

1. Linear Perceptron

The linear perceptron is a simple linear classifier that attempts to find a separating hyperplane between the two classes. It works well when the data is linearly separable. This model will serve as a baseline for comparison.

2. Logistic Regression

Logistic regression is also a linear classifier, but produces probability outputs for each class. It is generally more stable than linear perceptron and performs better when classes overlap. It also allows us to interpret how individual features contribute to predictions.

3. Support Vector Machine (SVM)

The SVM finds the optimal separating hyperplane by maximizing the margin between classes. SVM often performs better than other linear models because it focuses on the most important data points near the decision boundary. We expect SVM to achieve the best performance among the 3 models

All models will be trained using the same dataset splits. We will compare their performance using:

- Accuracy
- Precision
- Recall
- F1-score

This comparison will help us determine which algorithm is most effective for predicting sleep efficiency.

### 3.4 Expected Outcomes

We expect to observe relationships between lifestyle factors and sleep efficiency. For example, we predict that higher caffeine consumption, higher alcohol intake, smoking status, and increased awakenings may be associated with poor sleep efficiency. In contrast, regular exercise and balanced sleep composition may be associated with good sleep efficiency.

In terms of model performance, we expect:

- The linear perceptron to perform the worst if the data is not perfectly linearly separable.
- The logistic regression to perform better than the perceptron due to its probabilistic nature.
- SVM to achieve the highest accuracy because it maximizes the margin and generalizes better.

Through this project, we expect to better understand both the dataset and the strengths and weaknesses of each classification algorithm.